A cached-head 194-state machine for PA consistency

A verified composition of the PA194 decoder and the 195C controller

Research prepared for Vladimir Reshetnikov
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Abstract

We compare the PA-consistency machine families present in the requested ProveIt directories at commit `768c2be1a8e4d2fcf8aa3443c7b0f084d145734c`. The completed construction is a materially different binary, single-tape, blank-start machine with 194 working states. It combines the 67-reader, seven-fold AX decoder of PA194 with the cached-head Shift and modus-ponens controller of 195C. The combination changes six native transition rows, and implements a 3,649-site source program with 4,749 primitive outcomes. Both a construction checker and an independent literal-tape symbolic checker pass. The latter reconstructs its arithmetic-list lemmas from the actual rows, checks arbitrary register values, and checks the exact 57,922-step blank startup. A syntactic consistency calibration and positive finite-trace simulation establish the ordinary mathematical statement

$$\mathsf{PA} \vdash \big(\neg\,\mathrm{Halt}(\mathsf{PA194C}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathsf{PA})\big).$$

The contribution is this checked composition and its corpus comparison; the separate-head calculus, cached-head protocol, and decoder folds are credited to their supplied sources. No new end-to-end Lean or Rocq endpoint is claimed. The bounded reduction searches reported here do not establish a smaller machine or a lower bound.

Literal machine. Working states $0, \ldots, 193$; initial state 0; separate halt label 194; alphabet $\{0, 1\}$; a two-way infinite tape initially all zero. All 388 rows appear in Appendix A.

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1 Result and scope

The user requested fewer than 194 states, with a materially different 194-state construction as the fallback. The completed positive result is the fallback, denoted PA194C. Its count is

$$\boxed{134 \text{ fixed controls} + 67 \text{ raw AX readers} - 7 \text{ compatible folds} = 194.}$$

The fixed count already includes the initializer, tape arithmetic, leaf routing, and source controller. The halt label is not counted. There is no prewritten input, hidden extra tape symbol, stay-put transition, or unaccounted interpreter.

The dense transition table is `machine/pa194C.tm`. Its SHA-256 is

`d759313c03b515293fe06b12769fa1771d3829a2bad89febaa9b4391990f3a4b.`

The native table uses start 74, source-frame entry 84, and halt 254. Its SHA-256 is

`b35d80b01ce9c2e95c8b29222881f40928247cdeffac322e4643f4f6694cd33c.`

The supplied label bijection makes the dense and native tables exactly conjugate. The new graph differs from every recognized 194-state graph in the inspected corpus under arbitrary control renaming, a changed designated root, and global left/right reflection. Section 2 and its machine-readable inventory record the wider corpus check.

1.1 Three levels of evidence

The report distinguishes the mathematical theorem, executable certificates, and proof-assistant formalization. The mathematical proof uses syntactic proof translations and finite-trace induction inside PA. The supplied Python programs check literal table bindings, the source graph, symbolic physical contracts, local folds, and startup. Those programs are independently inspectable executable evidence. They are not an exported Lean/Rocq theorem of the complete PA biconditional.

This distinction matters in the source repository: its PA194 and PA195 iPRA endpoints retain machine-specific reductions as hypotheses, whereas its quotient developments prove ordinary finite-run statements. These are valuable proof layers for the same table, but do not by themselves close the PA-internal consistency equivalence [6, 7]. The supplied PA194 and 195C manuscripts also contain ordinary mathematical calibration arguments; their use here is explicit [1, 2].

2 Comparison of the existing designs

The comparison uses the revision stated in the abstract. A final freshness check incorporated two archives added after the initial snapshot. All 36 top-level ZIPs in `docs/incoming` are traversed, together with every direct numeric `.tm` table in the PA directory. Nested ZIP payloads are deduplicated by their bytes. The census recognizes complete numeric binary transition tables in `.tm` and `.txt` members. Tables encoded solely as arbitrary source code or JSON are outside that parser; the family review separately reads the reports and construction recipes. The inventory includes experimental and rejected machines, so its totals must not be read as a count of proved consistency constructions.

The fresh census contains 49 distinct ZIP bodies and 77 archive references, with maximum nesting depth 8. Across 26,258 members it recognizes 2,519 table occurrences, 606 byte forms, and 437 root-free control-graph classes. At 194 states there are seven literal occurrences, seven byte forms, and four graphs: principal PA194, PA194_AD, the further 68-reader/eight-fold PA194

allocation, and the rejected seven-zero scanner quotient. At 195 states there are 82 occurrences, 25 byte forms, and 15 graphs: five principal positive manuscript families, two retained search intermediates, two finite-verified secondary logical variants, and six rejected or diagnostic classes. At 196 states the counts are 788, 163, and 99. Two disconnected synthetic controls are excluded from graph classification.

The new graph is absent from *all four* recognized 194-state classes, including the rejected one, and from each global tape reflection. All seven old tables are included in `comparison/prior194/`; the replay is `comparison/check_all194_novelty.py`. The complete member-level inventory is `comparison/census.json`.

2.1 The relevant frontier

Family	States	Fixed/raw/folds	Source sites	Outcomes	Startup
Original PA195	195	134/68/7	3,700	4,815	58,154
195S	195	134/67/6	3,673	4,780	58,154
195C	195	134/67/6	3,649	4,749	57,922
PA195DC	195	134/68/7	3,676	4,784	57,922
M195S8	195	135/68/8	3,700	4,815	59,164
PA194	194	134/67/7	3,673	4,780	58,154
PA194_AD	194	134/67/7	3,673	4,780	58,122
PA194 $(68-8)$	194	134/68/8	3,673	4,780	58,154
PA194C (this work)	**194**	**134/67/7**	**3,649**	**4,749**	**57,922**

Original PA195 and M195S8 use the bundled, 191-token diagonal-vacuity emitter. The latter spends an extra scanner control to validate an eight-fold allocation. The 195S/C and PA194 families instead use the 190-token separate-head emitter. The original PA195 has seven folds but 68 raw readers; 195S has 67 raw readers but six folds. PA194 attains both savings together. The present construction keeps that decoder and uses the 195C source controller. Its source is 24 primitive sites and 31 outcomes shorter than PA194. This is a source-level protocol change, not a choice of finite-control labels.

The later PA194 packet supplies another allocation of the same separate-head source, with 68 raw readers and eight folds. Its source controller is the original one; its graph and reflection also differ from the delivered cached-head machine. The later PA195DC combines the 191-token diagonal emitter with the cached controller at 195 states. Thus it shares the protocol change, but uses a different emitter and 3,676 source sites. Both additions are included in the refreshed census and archive ledger [3, 4].

2.2 Larger families, logical changes, and physical cost

The larger tables separate into ancestry, supported alternative implementations, and diagnostics. The following taxonomy summarizes the supplied mathematical arguments and implementation evidence; it does not infer calibration from a state count. Source keys refer to the exact archive locators below and their entries in Appendix B.

Family and size	Change, evidence, and cost driver
PA209 and U209 209 states	James Chang's PA209 has 3,948 primitive source sites; the U209 ancestor used by the U208 packets is a different 3,909-site construction. Their equal counts do not identify their source or their physical controls. The published PA196 lineage already changes the quantifier encoding, equality interface, head order, and contextual sharing [7, 1].
U208 208 states	A finite-history quotient identifies old roles 172 and 211. Alternative helpers give 20 designated, supported graph classes among 27 presentations; this is a smaller set than all experimental 208-state graphs in the census. The packets give local halting-equivalence arguments and inherited calibration. Repeated entry/context/history proofs describe the same one-state saving [L1].
Two different 207 stories	The U208 attempts include rejected 207-state frame witnesses; they are not successful successors to U208. Separately, a repaired 157-node representation of the PA196 source has a verified 207-state realization: 3,812 sites, 4,959 contracts, and exact startup. Lost contextual helpers and additional routing make that smaller source DAG physically larger [L2,L3].
Integrated witness gate $206 \to 203$ states	A logical gate substitution removes 67 source sites, giving 3,745. Both the initial 206-state and compressed 203-state realizations pass 4,872 source contracts and full startup checks. The packet supplies the logical substitution argument, but its reader/routing allocation still costs more than PA196 [L4].
Published PA196 and same-size physical forks	The baseline has 3,812 sites and $36 + 13 + 147 = 196$ controls for the engine, leaf routing, and readers. Reassociation, head ordering, A/D register exchange, closing-decrement choices, and zero-jump routing produce further 196-state implementations with distinct hypotheses. Several closing variants edit the same row, so their savings cannot be added. The explicit zero-jump invariant is another separate obligation [L5].
Gate/quantifier forks 200, 199, 198, 196	The first diagonal-vacuity implementation has 3,700 sites and 199 states; later association gives 198 and then 196. Distributed-gate compilation has 3,773 sites and 198 states. Combining shorter negation and changed MP with different gates yields additional 196–200 tables. The packets separate finite implementation checks from the logical and command translations needed for each composition [L6].
Short negation 197 states	A direct-negation realization has 3,556 source sites. The later NEG18 macro reduces a separate-emitter source to 3,441 sites and 4,468 outcomes, yet its final allocation is $134 + 68 - 5 = 197$. Its arithmetic macro and scratch-register invariant are checked; losing useful shared readers prevents the source saving from becoming a state saving [L3,L7].
Changed Shift/MP mostly 196 states	H-only Shift alone has 3,811 sites. Combined Shift/MP has 3,788 sites, 4,928 uniform contracts, and exact startup. Crossing this controller with diagonal/separate emitters gives 3,676/3,649 sites, initially at 196 states. Those fully checked constructions differ from earlier MP-only experiments supported merely by bounded frame evidence; two rejected Shift layouts actually halt at step 43,563 [L3,L4,L8].
Diagnostics and incomplete components	A runtime-checked 198-state pruned source lacks its calibrated blank startup; a parity-control component also has 198 states without a complete endpoint. Short-scanner and optimistic reader quotients fail specified frame contracts. Such tables remain in the census for comparison, without being counted as established consistency machines [L3,L5 and the rejected-frontier discussion].

Source locators. All following ZIPs are under `docs/incoming/` at the report's pinned revision. L1: `U208_ConPA_2026-10-29.zip` and `pa208_research_2026-10-29.zip`; L2: `u208_comparison_research_2026-10-29.zip`; L3: `conpa_tm_study_2026-11-13.zip`, member `conpa_tm_study/report.tex`; L4: `pa_consistency_reduction.zip`, member `pa_consistency_reduction/report.tex`; L5:

`pa_tm_research_2026-11-13.zip`, `pa_tm_extended_research_2026-11-13.zip`, `conpa_tm_comparison_and_search.zip`, and `pa_tm_research_2026-11-13(2).zip`; L6: `pa_consistency_research_2026-11-13.zip`, its `(2).zip` companion, `pa_tm_comparison_and_reductions.zip`, `pa_consistency_comparison_2026_11_13.zip`, and `pa_consistency_comparison_and_new196_2026-11-13.zip`; L7: `pa_consistency_research_2026-11-14.zip`; L8: `pa_consistency_cross_archive.zip` and `conpa_tm_comparison_and_reduction_search.zip`. Their hashes and every recognized table occurrence are in the ledger and `comparison/census.json`. These families explain why logical strength, source size, physical sharing, and initialization must be compared separately before a proposed reduction is accepted.

2.3 Rejected constructions constrain the search

Several numerically smaller tables fail their intended source simulation. The earlier two-rotation 195 quotient has 93 recorded primitive-contract failures. Another 195 regrouping introduces a seven-zero source address that the six-zero scanner cannot handle. The follow-up packet similarly contains a rejected 194 table with valid local folds and correct source graph but an invalid physical scanner. Adding a scanner state repairs that particular issue at a higher count. Two older rejected Shift layouts actually halt on blank tape after 43,563 steps. A table with a smaller row count is consequently only a candidate until its source interpretation and blank initialization are checked [1, 5].

A failed certificate does not prove that the same table could have no other interpretation with the desired blank-tape property. It does show why the supplied failed construction cannot be counted as a success. Likewise, negative search results apply to their stated rewrite families, not to all Turing machines.

3 The construction

3.1 The six physical edits

Use PA194's native labels and its unchanged seven-fold decoder. Replace exactly the rows in the following table; every other row remains identical to the primary PA194 native table. Each triple is "write, direction, next control."

State	Read	PA194 action	PA194C action
38	0	0R28	0R11
38	1	1R20	1R122
164	1	1R98	1R130
214	0	0R188	0R312
214	1	1R312	1R20
312	0	0R11	0R188

The corresponding source replacements are

$$\texttt{node163} := \mathrm{node}(\texttt{node6}, \texttt{node147}),$$
$$\texttt{node170} := \mathrm{node}\big(\mathrm{node}(\texttt{node36}, \mathrm{node}(\texttt{node167}, \texttt{node145})), \texttt{node57}\big).$$

All six rows and both replacements come from the existing 195C construction. Their composition with the PA194 decoder is new relative to the inspected corpus. They implement Shift's head-only contradiction test and return a successful MP result in the cached head.

The builder reconstructs both the new machine and the published 195C parent. The latter's native SHA-256 is

`8f34d9e5cc2e64ecc49bbc8581dc29b2ac0a84ad9187c4dd8b7597588ef22ea3.`

Both resolved source instruction maps equal the independently supplied 195C instruction map, of SHA-256

`0afee1346c0931b3b91fea70571b01fd665a802ab9327409e4030f673412f5e3.`

Thus the construction is bound to an existing calibrated source program, not merely to a visually similar macro word.

3.2 Decoder and simultaneous folds

The AX emitter keeps the exact 190-token separate-head macro word. Its binary grammar has 67 allocated raw reader states. The seven native redirects are

$$2016 \mapsto 2052, \quad 2018 \mapsto 2027, \quad 2022 \mapsto 2032,$$
$$2024 \mapsto 2046, \quad 2037 \mapsto 2015, \quad 2055 \mapsto 2019, \quad 2060 \mapsto 2014.$$

Their retained rows are exactly those in the delivered `construction/source/pa194.folds.json`. For each pair, one old control is entered after writing 0 and moving right; the other is entered after writing 1 and moving right. The bit immediately to the left therefore records which old control was intended. A short left/right detour through existing readers recovers that bit and simulates the appropriate old transition.

It is essential to check all seven folds in the final simultaneous table. A helper used by one fold may itself be changed by another. The checker exhausts the two history bits and two current symbols for each fold, giving 28 cases. In each case, the final control, head position, and the entire two-cell window agree with the original single transition after control projection. No other cell is touched. The incoming-row checks justify the stored history bit; the start is not a merged control.

Lemma 3.1 (Positive local quotient)**.** *The folded table simulates each raw transition from an admissible entry by a finite positive number of physical transitions, preserving the complete tape and the projected endpoint.*

Proof. An unchanged row commutes with the projection of targets. For a changed row, the incoming context selects one of the four exhaustively checked two-cell cases. A two-cell calculation is valid on every extension of that window to a complete tape because its trace never reads outside the window. The checked endpoint and positive duration prove the claim. Induction concatenates these simulations along finite raw runs. □

The quotient lemma explains the sharing mechanism. The independent whole-source check below also validates the literal final folded table, so the physical conclusion does not depend solely on the allocator's implementation of this argument.

4 The cached-head command protocol

This section makes the semantic difference concrete and records the proof compiler that justifies it [2]. Define Cantor pairing

$$p(a, b) = \frac{(a + b)(a + b + 1)}{2} + b, \qquad [a; w] = p(a, w).$$

The total projections satisfy $\mathrm{hd}[a;w] = a$ and $\mathrm{tl}[a;w] = w$, also when $a = w = 0$, where $[0;0] = 0$. Use $I(A,B) = 4p(A,B)$ and $U(a,B) = 4p(B,a) + 2$. A command store is $\langle t, w, d, h\rangle$: current formula, coded stack, residual counter, and cached head.

The new commands are

$$\mathsf{S}\langle t,w,d,h\rangle = \begin{cases} \mathsf{halt}, & h = 1, \\ \langle h, \mathrm{tl}(w), d, \mathrm{hd}(w)\rangle, & h \neq 1, \end{cases}$$

and, writing $y = d + x$,

$$\mathsf{MP}_x\langle t,w,d,h\rangle = \begin{cases} \langle 0,w,0,y\rangle, & I(t,y) = h, \\ \langle 0,w,y,0\rangle, & I(t,y) \neq h. \end{cases}$$

Only Shift halts. A successful MP leaves the stack untouched and writes the conclusion into h. This is why the unchanged old command compiler cannot simply be assumed to work.

4.1 AX, stack operations, and zero aliases

The separate-head AX command pushes ten entries including the saved top:

$$\mathsf{AX}_{c,x,y}\langle t,w,d,h\rangle = \langle K(h+y, d+x, c), [q_1,\ldots,q_9,t;w], 0, 0\rangle.$$

Here $q_1 = \Gamma(c)$ is the separate diagonal head, the next six entries are the six ordinary logical heads, q_8 is the dominated implicational head, and $q_9 = U(d,h)$. The top K is the guarded gate, or zero if its numeric guard fails. None of $K, q_1, \ldots, q_9$ equals 1: the gate and the first eight heads have residue 0 modulo 4, and q_9 has residue 2.

The projection identities give two uniform, nonhalting computations:

$$\mathsf{AX}_{c,x,y};\mathsf{S}^9 : \langle t,w,d,h\rangle \longmapsto \langle q_8, [t;w], 0, U(d,h)\rangle,$$
$$\mathsf{AX}_{c,x,y};\mathsf{S}^{10} : \langle t,w,d,h\rangle \longmapsto \langle U(d,h), w, 0, t\rangle.$$

These words do not test the saved t, even when $t = 1$. Define

$$\begin{aligned} \mathsf{Push} &= \mathsf{AX}_{0,0,0};\mathsf{S}^9;\mathsf{MP}_0, \\ \mathsf{Pop} &= \mathsf{AX}_{0,0,0};\mathsf{S}^{11}, \\ \mathsf{Discard} &= \mathsf{MP}_1;\mathsf{AX}_{1,0,0};\mathsf{S}^{10};\mathsf{MP}_0. \end{aligned}$$

Their contracts are

$$\begin{aligned} \mathsf{Push}\langle t,w,d,h\rangle &= \langle 0, [t;w], 0, 0\rangle, \\ \mathsf{Pop}\langle t,[a;w],d,h\rangle &= \begin{cases} \mathsf{halt}, & t = 1, \\ \langle t,w,0,a\rangle, & t \neq 1, \end{cases} \\ \mathsf{Discard}\langle t,w,0,0\rangle &= \langle 0,w,0,0\rangle. \end{aligned}$$

Push's final MP fails because an implication has residue 0 and the cached U has residue 2. Pop's last Shift tests precisely the preserved top. For Discard, $I(t,1) > 0$ forces the first MP to fail; after AX and the ten Shifts, $I(U(1,0),0) > 0$ forces the last MP to fail. These proofs include the zero-stack alias and require no fictitious stored stack length.

4.2 A proof compiler for the changed protocol

Here are all selector families for the ten-entry layout, as proved in `reference195C/independent_verification_composite_command_audit.tex`. For $1 \le j \le 7$, the word

$$\mathsf{Discard};\mathsf{AX}_{A,D,H};\mathsf{S}^{j+1};\mathsf{Pop}^{9-j}$$

returns q_j with the original underlying stack and zero caches. The Shifts retrieve q_j; the Pops preserve it while discarding the remaining generated entries and the saved zero. These Pops cannot halt because $q_j \neq 1$. Define Offset_0 to be empty and $\mathsf{Offset}_u = \mathsf{MP}_u$ for $u > 0$. For $u \le a$, the word

$$\mathsf{Discard};\mathsf{Offset}_u;\mathsf{AX}_{b,a-u,c};\mathsf{S}^9;\mathsf{Pop}$$

returns the dominated head $\rho_8(a,b,c,u)$. The gate selector is

$$\mathsf{Discard};\mathsf{AX}_{c,b,a};\mathsf{Pop}^{10},$$

which returns $K(a,b,c)$, including zero on a failed guard. All three contracts start with arbitrary top and stack and zero caches, and restore the stack and zero caches. In particular, the diagonal head is selected by

$$\mathsf{Discard};\mathsf{AX}_{B,0,0};\mathsf{S}^2;\mathsf{Pop}^8.$$

It yields $\Gamma(B)$ with the underlying stack restored and both caches zero, even when the gate guard fails. The generalization selector is

$$\mathsf{Gen}_i = \mathsf{Push};\mathsf{Offset}_i;\mathsf{S};\mathsf{AX}_{i,0,0};\mathsf{S}^{10},$$

It sends $\langle a,w,0,0\rangle$ to $\langle U(i,a),w,0,0\rangle$, including $a=1$.

The critical MP suffix is

$$\mathsf{MPSuffix}_b = \mathsf{S};\mathsf{S};\mathsf{MP}_b;\mathsf{S}.$$

It acts on $\langle 0,[a,I(a,b),0;w],0,0\rangle$ by halting when $a=1$ or $b=1$, and otherwise returning $\langle b,w,0,0\rangle$. The first Shift retrieves a, the second tests it and retrieves $I(a,b)$, MP writes b into h, and the last Shift tests b and retrieves the dummy zero. Omitting that dummy zero would consume an entry of w. The same numerical calculation works when the displayed zeros alias the empty coded stack.

Compile an MP proof node with minor premise a, major premise $I(a,b)$, and premise proof trees p_0,p_1 by

$$W_p = \mathsf{Discard};\mathsf{Push};W_{p_1};\mathsf{Push};W_{p_0};\mathsf{Push};\mathsf{MPSuffix}_b.$$

Axioms use the selectors and generalization uses $W_{p_0};\mathsf{Gen}_i$. Induction on proof trees gives the following uniform invariant.

Lemma 4.1 (Proof-word completeness)**.** *For every proof tree p of a numerical formula A and every t,w, W_p from $\langle t,w,0,0\rangle$ either halts or returns $\langle A,w,0,0\rangle$. If p proves 1, the word*

$$\mathsf{AX}_{0,0,0};W_p;\mathsf{Pop}$$

halts from every incoming command store.

Proof. The head selectors and generalization contract give the corresponding induction cases. In the MP case, Discard and the first Push create the dummy zero. The two premise computations and their Pushes establish the suffix precondition; the suffix either halts or returns the conclusion. An earlier halt is preserved by extending the word. The initial AX in the displayed refutation word clears d,h, regardless of the incoming top and stack. If its compiled proof has not already halted, Pop tests the final top 1 before relying on the arbitrary underlying stack. All operations and proof-tree constructions are primitive recursive. □

The inherited exhaustive command-word scheduler need not reset its store between trials: the refutation word succeeds from every incoming store. Its finite enumeration and decoding lemmas give an explicit finite scheduler segment containing that word. This is the completeness property used in the consistency argument, not a claim that arbitrary trials leave the store empty.

4.3 Binding the changed primitive loops

The changed command tails have direct all-natural loop invariants. In the 3,649-site source, the successful MP tail starts at ordinal 3,634. It transfers $(D, H) = (y, 0)$ to $(0, y)$ by a decrement/increment/return loop, taking $3y + 1$ primitive instructions to exit and one more to return to fetch. The failure branch leaves $(D, H) = (y, 0)$ and returns directly. The new Shift starts at ordinal 3,638. Under its inherited scratch-entry condition $X = 0$, it clears T, copies H into T, X, and restores H from X. It reaches $(T, H, X) = (h, h, 0)$ in $2t + 7h + 3$ instructions, then sets $X = 1$ and enters the unchanged guard and Cantor decoder. These elementary loop calculations bind the stated command protocol to the two altered source subtrees. The full source-graph comparison and physical checks bind them further to the literal machine.

5 PA-provable consistency calibration

The calibration uses the same separate-head numerical calculus as 195S/C and PA194. We recall its proof-producing content, because an external statement that the machine never halts would not suffice for the user's requested PA theorem [1, 2].

5.1 Numerical syntax and the diagonal clause

In addition to I, U, set $M(a, b) = 4p(a, b) + 1$, $E(a, b) = 4p(a, b) + 3$, and $N(B) = I(B, 1)$. Zero is the designated truth code, one is $M(0, 0)$, and the U arguments are transposed as defined above. Conjunction and existential quantification have their classical implicational definitions. The separate diagonal clause is

$$\Gamma(B) = I(B, U(B, B)).$$

All variable labels occurring in the decoded formula B are strictly below B. Thus x_B is fresh in its own formula. Structural descent on codes proves this, treating truth code 0 separately; the constructor alias $I(0, 0) = 0$ is not an ordinary disjoint syntax case.

Write

$$\begin{aligned}
V(a, b, c) &= M(a, b) \to U(c, M(a, b)),\\
P(a, b, c) &= c \to M(c, b),\\
S(a, b, c) &= U\big(b, (M(a, b) \to P(a, b, c)) \wedge (M(c, b) \to c)\big),\\
Q(a, b, c) &= U(c, S(a, b, c) \to N(P(a, b, c))),\\
G(a, b, c) &= N(V(a, b, c) \to Q(a, b, c)).
\end{aligned}$$

The AX top is $G(a, b, c)$ when $c \neq a, b$, and zero otherwise; the separate $\Gamma(c)$ is emitted without

that guard. The six other logical heads are unchanged from the supplied PA calculus:

$$\begin{aligned}
&(\exists x_a\, U(a, C)) \to C,\\
&U(a, U(b, C)) \to U(b, U(a, C)),\\
&M(c, a) \to (E(c, b) \to M(b, a)),\\
&E(c, a) \to (M(b, c) \to M(b, a)),\\
&\exists x_a E(c, a),\\
&U(a, b) \to \big(U(a, b \to c) \to U(a, c)\big).
\end{aligned}$$

The dominated implicational head is

$$\rho_8(D, A, H, d) = ((H \to d) \to D) \to ((D \to H) \to (A \to H)), \qquad d \le D.$$

The inference operations are modus ponens and generalization of theorems. The free-algebra emitter audit retained with the package fixes the exact parameter order and stack order of these heads.

The bundled gate $N(\Gamma(c) \to (V(a, b, c) \to Q(a, b, c)))$ and the pair $\Gamma(c), G(a, b, c)$ are interderivable by fixed classical implicational proof templates. Every diagonal instance $\Gamma(B)$ is available from a bundled gate by taking $c = B$, $a = b = B + 1$. Consequently the bundled and separate-head presentations have primitive-recursive proof translations.

5.2 PA refutations yield numerical refutations

The inherited PA-refutation compiler uses two-way membership rewriting, neighborhood equality, and an equality witness. The supplied diagonal calculus reconstructs the interfaces that formerly used equality-vacuity. For a reverse rewrite, put

$$C = E(a, b) \to (\chi(b) \to \chi(a)), \qquad \varphi = N(C), \qquad c = \ulcorner \varphi \urcorner.$$

Here χ is a membership atom with the indicated argument in either slot, or the diagonal membership case. Code descent provides freshness. Pointed separation, the retained equality witness, direct membership rewriting, and membership-only vacuity derive $\varphi \to (S \to N(P))$. Generalization at c, discharged using $\Gamma(\varphi)$, gives $\varphi \to Q$, while the separation instance provides $N(Q)$. Classical double-negation elimination gives C. Coincident labels are handled by the corresponding propositional cases.

For neighborhood equality, first use the fresh label $z_0 = \ulcorner E(a, b) \urcorner$, the instance $\Gamma(E(a, b))$, and the two-way rewrites. The equality witness and membership-only vacuity then transport the binder to any required $z \neq a, b$. These are finite proof templates with primitive-recursive code construction and label comparisons. They supply the original compiler interfaces, and their correctness can be proved in PA by induction on the source proof. Together with Section 4 and the scheduler, this gives a PA-verifiable translation from a PA refutation to a source halt.

5.3 Source halts yield PA refutations

Translate $M(a, b)$ to finite binary membership $\mathrm{Bit}(x_a, x_b)$, $E(a, b)$ to equality, and $U(a, B)$ to universal quantification of the translated formula. One arithmetic definition is

$$\mathrm{Bit}(s, z) \equiv \exists u, v\, [s = (2u + 1)2^z + v \ \wedge \ v < 2^z].$$

Exponentiation can be given its standard definitional arithmetic representation. PA proves $\mathrm{Bit}(s, z) \to z < s$, hence the negation of the translation of code 1. The alias $I(0, 0) = 0$ is handled by a fixed proof of truth's equivalence to truth implying truth.

For each arithmetic formula $\theta(z, \vec{u})$, PA proves finite binary comprehension:

$$\forall n \, \exists s < 2^n \, \forall z < n \, (\mathrm{Bit}(s, z) \leftrightarrow \theta(z, \vec{u})).$$

Induct on n, retaining the previous code when $\neg\theta(n, \vec{u})$ and adding 2^n when $\theta(n, \vec{u})$. Substitution of the syntax of θ into this fixed induction-proof pattern is a primitive-recursive proof supplier, whose correctness PA verifies syntactically. This uses neither a truth predicate for arbitrary arithmetic formulas nor a reflection principle for PA.

For the separation gate, the finite matrix when $a \neq b$ is

$$(\mathrm{Bit}(x_a, z) \vee z = x_b) \wedge T(c)[z/x_b],$$

with a bound such as $x_a + x_b + 1$. When $a = b$, the correct matrix is $(z = x_b) \wedge T(c)[z/x_b]$. The binder then changes both occurrences of the coincident label, and its other antecedent is $\mathrm{Bit}(z, z)$, which PA refutes. Keeping this collision case separate avoids an invalid free-variable argument. The remaining heads have fixed first-order proof suppliers, and the diagonal head uses the proved code-freshness fact.

Annotate the current formula t, cached head h, and every live canonical stack entry with a PA proof of its translation. The canonical stack representation explicitly handles $p(0, 0) = 0$: popping coded zero supplies the truth proof, and pushing zero onto zero creates no spurious live cell. AX supplies the head proofs. Successful MP combines the proofs of t and $I(t, y)$ and places the resulting proof with the new cache y. Failed MP supplies truth proofs for the cleared formulas. Shift moves the annotations and can halt only at $h = 1$, which therefore carries a proof of the translated contradiction. Induction over any finite command trace constructs a PA refutation from a halt.

Proposition 5.1 (Syntactic source calibration)**.** *Let R_C be the cached-head source program, with its exact blank-derived initial store. The proof constructions above are arithmetizable in PA and give*

$$\mathsf{PA} \vdash \big(\mathrm{Halt}(R_C) \leftrightarrow \exists p \, \mathrm{Prf}_{\mathsf{PA}}(p, \ulcorner 0 = 1 \urcorner)\big).$$

Proof. The forward implication is the proof-annotation transformation. The reverse implication is the PA-refutation translation, the corrected cached-head proof-word compiler, and the exhaustive scheduler. These are syntactic transformations of finite objects. PA proves their correctness by the induction and finite arithmetic arguments described above. In particular, neither direction assumes that PA is sound or consistent. □

6 Physical correctness and the final theorem

6.1 Universal source frames

Each source site has a binary path p. Its clean tape frame contains that path, a following zero gap of length $48 + n - |p|$, a marker, nine unary register blocks, and a finite nonempty list of spare blocks. Here n and the nine register values are independent arbitrary natural numbers, and the spare-list values are arbitrary naturals. A register value r is represented by a block of $r + 1$ ones following its separator.

The logical-to-physical register permutation is

$$(6, 2, 0, 4, 3, 7, 8, 5, 1).$$

It is unchanged from the principal PA194. All increment outcomes and both decrement outcomes are verified, together with jumps and halts. The independent source resolver computes the graph by

two methods and checks equality with the published 195C instruction list. Merely keeping the leaf word would not have justified relative break/continue targets.

The independent checker reconstructs four generic list-sweep lemmas directly from the supplied literal rows. It then symbolically executes every source outcome. Durations are nonnegative affine expressions in the symbolic sizes with strictly positive constants. Thus these checks cover arbitrary register values and spare-list lengths, rather than a finite sample of concrete runs.

Quantity	Construction checker	Independent checker
Source instruction sites	3,649	3,649
Source outcomes	4,749	4,749
Symbolic execution steps	1,507,397	1,096,302
Maximum symbolic steps for one outcome	506	403
Arithmetic-list lemmas rederived from rows	–	4
Minimum positive-duration constant	–	3
Blank startup transitions	57,922	57,922

The two symbolic-step totals count different implementations' rewrite operations; neither is a claim about a uniform physical running time. The construction checker uses the unchanged imported arithmetic-block interfaces. The independent checker uses no arithmetic-box oracle.

6.2 Exact blank initialization

The full blank run reaches native state 84, head position -2, at source site 25 after 57,922 transitions. The logical register vector is $(2, 0, 0, 0, 0, 0, 0, 0, 0)$, with 26 spare values equal to 1 and header distance 48. Both checkers compare the entire tape, including its zero exterior, with the prescribed source frame. One uses a finite-support set and the other a bounded byte array with explicit bounds checks. The complete nonzero support and source path are recorded in `verification/independent/recorded_C/PA194C/blank_seed.json`.

The startup is 232 transitions shorter than that of principal PA194. This is a startup comparison only; it does not claim uniform speedup on all subsequent computations.

Lemma 6.1 (Finite-trace transport inside PA)**.** *PA proves that* PA194C *halts from blank tape exactly when R_C halts from the specified source seed.*

Proof. The exact finite startup establishes the first source frame. Every source instruction has a checked, positive-duration physical simulation to the next frame or to halt. The finite symbolic certificates are equations and elementary list/arithmetic inductions; their durations are primitive recursive in the frame data. PA can verify the finite table equalities and prove these frame lemmas by the indicated inductions. Induction on source length concatenates them, proving that a source halt produces a physical halt.

Conversely, suppose a physical halt occurs at time T. If the first $T + 1$ source steps were all nonhalting, their simulated endpoint times would be strictly increasing natural numbers. They would therefore carry the unique deterministic physical run beyond time T without halting, a contradiction. The necessary source prefixes and durations are finite, primitive-recursively coded objects. This bounded argument is provable in PA and avoids an appeal to an external infinite run. The fixed native/dense conjugacy transfers the statement to the delivered labels. □

Theorem 6.2 (The delivered PA consistency machine)**.** *For the literal dense table in Appendix A,*

$$\boxed{\mathsf{PA} \vdash (\neg\,\mathrm{Halt}(\mathsf{PA194C}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathsf{PA})).}$$

Proof. Compose finite-trace transport with the syntactic source calibration, then negate the existential PA-refutation predicate. This is an ordinary mathematical proof of PA-provability. The supplied executable checks bind its finite data to the literal table; an end-to-end proof-assistant export is further work. □

7 Search for fewer states

Three additional grammar searches ran for 600 seconds each. The target was at most 59 effective AX readers with zero scanner penalty, which would give at most 193 working states with the fixed 134-control component. All runs began at the accepted 67-reader, seven-fold PA194 grammar. The best penalized cost remained 60 in every run; no target hit occurred.

Run	Controller and mutation	Proposals	Accepted moves
A	Original; global DAG rotation	10,877,569	672,063
B	Original; global and occurrence rotation	9,328,284	1,747,096
C	Cached-head; global and occurrence rotation	9,521,809	1,673,285

The three seeds were respectively 2026111405, 2026111406, and 2026111407. Counts include repeated proposals and do not count distinct machines. The runs keep the emitter word and parameter roles fixed. The source, exact numerical inputs, completed logs, and a receipt are supplied in `search/grammar/`. Wall-clock limits make proposal counts dependent on the computer. No scored grammar is accepted as a machine without the independent source and physical checks.

The freshness review added two further searches from the newly supplied 68-reader/eight-fold allocation, using the original and cached-head controllers and seeds 2026111408 and 2026111409. Both started at cost 60. Their 600-second budgets were interrupted before a final completion record. The last complete progress records contain 11,512,000 proposals at 571.320 seconds and 11,484,000 at 571.352 seconds, respectively. Neither recorded an improvement or target hit. These are partial-run counts, not claims that the budgets completed. The exact inputs, source and starting-frame audits, partial logs, and bounded local searches are retained in `search/grammar/fresh_frontier/`. The additional local checks used radius 4 with duration 120 and radius 8 with duration 200 for both starting controllers; they returned no candidate.

Whole-source quotient searches tried a further 460,386 candidate tables, executing 1,580,604 symbolic contracts before rejection or a resource bound. No candidate passed the preliminary source-frame check.

Machine	Replacement-action family	Candidate tables	Contracts
PA194	Direct two-row choice	57,460	245,263
PA194C	Direct two-row choice	57,460	244,771
PA194	Additional valid local detours	171,009	541,573
PA194C	Additional valid local detours	174,457	548,997

The direct family merges pairs among 170 non-box, non-dispatcher controls and chooses each retained read-bit action from one of the two old states. Each source contract has a 1,024-symbolic-step limit. Counts include duplicate constructions. Although arithmetic controls are not merged, renaming a target can change a box's outward edge; any positive result would still require fresh list-lemma, physical, and blank-start checking. These trials yielded no positive preliminary result.

Local window searches and incoming-edge elimination also yielded no candidate at radii 4, 6, and 8 with duration bounds 120, 160, and 200, respectively. The primary and cached-head tables were both covered, with an additional radius-8 check of PA194_AD. A source-context radius-4 screen was also performed. Detailed bounds and method limitations are recorded in `search/local/SEARCH_NOTES.md`.

The obvious scanner shortcut, removing native state 34 and sending $(23, 0)$ directly to 0L75, fails at source instruction 0. The source has 25 paths containing six consecutive zeros. Thus this shortcut cannot simply be combined with the existing source frame. None of these negative searches proves a lower bound or proves all attempted tables semantically incorrect: it reports failure to certify a reduction within specified transformation families and checking bounds.

The grammar searches begin from the actual accepted PA194 grammar. For promising scores, their cost includes the maximum simultaneously compatible set of history folds within the implemented two-cell family. The original controller and the cached-head controller are searched separately, because their fixed rows provide different possible helper detours. Global DAG reassociations preserve sharing; a second mutation mode rotates individual occurrences and can introduce a different sharing pattern. Scanner penalties account for maximum zero runs and transient increment paths. Any promising numerical score still requires exact source-graph and universal physical verification.

The local-control searches allow existing-control detours and include arithmetic controls in their broad finite-window screening. A separate incoming-edge transformation attempts to remove a control by replacing its incoming rows with simulated two-step blocks, while avoiding every edited row. The whole-source quotient search is complementary: it checks proposed quotients at source boundaries rather than requiring a short local window. Each scope and step bound is recorded in the search receipts; none is a general minimality result.

8 Ideas transferred from ZFC and concise iPRA

The useful transfer from the ZFC work is methodological and physical: source reassociation, use of already written marker bits as finite-control memory, simultaneous validation of contextual folds, and whole-tape symbolic contracts. The independent source-language and physical-checking components retain their upstream K225/K226 and U214 provenance. The ZFC repository itself emphasizes that short source expressions do not alone price a complete initialized machine [8].

One cannot substitute a ZFC-consistency machine for a PA-consistency machine just because its compiler is compact: the calibration target is different. Likewise, the concise-iPRA project measures complete checker implementations and provides unusually economical calculi, but a small iPRA checker is not automatically a small PA-refutation search machine. Its parser, term evaluator, rule dispatch, arithmetic representation, enumeration, and initialization must all be priced in the binary tape model [9]. The present construction retains the established PA calibration and applies the reusable compiler ideas where their physical cost can be checked.

9 Reproduction, limitations, and further questions

9.1 Reproduction

Python 3.10 or later with its standard library is sufficient for the delivered construction and both executable checks. Run from the package root, with assertions enabled:

```
python3 -B verify_release.py
python3 -B construction/build194C.py
python3 -B verification/independent/check194C.py
```

The first command checks the shipped hashes and stable receipt fields. The second reconstructs the machine and its 195C parent; the third checks the literal output independently. Regenerating evidence may rewrite receipts, so run the integrity check before editing the package. No network connection or repository checkout is needed for these checks. The optional search tools require C++17 and are separate from verification.

Rebuild the report with

```
python3 report/build_report.py
```

from the package root. This runs pdfLaTeX three times; a standard TeX Live installation is required. Source references and imported licenses are retained.

9.2 Precise limitations

The completed machine uses 194 working states. The work establishes no strictly smaller calibrated machine and no lower bound at 194. Its novelty is relative to the pinned supplied corpus, not a claim about all unpublished designs. The mathematical PA theorem depends on the explicit inherited numerical-calculus and scheduler arguments, together with the physical transport proved here. The source repository's open iPRA endpoints remain open; no claim about them is implied by this ordinary PA proof.

9.3 Further research questions

1. Can a scanner-valid allocation achieve at most 59 effective AX readers with 134 fixed controls? For example, $66 - 7$ or $67 - 8$ would yield 193 states. Compatibility must be checked after all folds are applied.

2. Can a controller-specific source invariant justify a reduction that the arbitrary-tape window abstractions necessarily miss, while retaining an elementary inductive certificate?

3. Can the cached-head protocol share a return path or a leaf router with an AX reader, saving a fixed control without losing a useful fold?

4. Can the guarded separation emitter use a cheaply constructed fresh label exceeding all parameters and eliminate its freshness guard? A logical reformulation is useful only if the repeated construction and storage of the new label costs fewer physical controls.

5. Can the successful frame certificates and syntactic proof suppliers be turned into a single compiled PA-internal Lean endpoint for this exact table? The intermediate quotient theorem and a conditional arithmetic endpoint should remain distinct until those obligations are closed.

A Complete dense transition table

Start at state 0, head 0, on an all-zero two-way tape. An entry $w\mathrm{D}q$ writes w, moves one cell in direction D, then enters q. H denotes the distinct terminal label 194, with no outgoing rows. The table below contains all 388 transitions.

q	read 0	read 1	q	read 0	read 1
0	1R1	1R2	97	0R36	1R112
1	1R3	1R1	98	0R111	1R75
2	0R0	1R2	99	0L63	0R109
3	0R4	0R1	100	0L62	1R113
4	1R5	1L6	101	0L90	1R114
5	1L7	1L8	102	0R115	0L14
6	0L7	1L6	103	0R69	1R103
7	0L9	1L6	104	0R116	1R95
8	0L10	0L11	105	0R116	1R67
9	1R12	0L9	106	0R117	0L14
10	0L9	0L10	107	0R118	1L71
11	1L8	1L11	108	0R119	1R120
12	0R12	0R13	109	0R14	1R109
13	0L14	0R13	110	0R66	1R110
14	1L15	0L15	111	0R121	1L35
15	0L15	1L16	112	0R122	1R123
16	0L17	1L16	113	0R88	0L14
17	0L18	1L16	114	0R90	1L95
18	0L19	1L16	115	0R124	1R124
19	0L20	1L16	116	0L63	0R103
20	0L21	1L16	117	0R68	0R91
21	0L22	1L16	118	0R58	0R91
22	0L23	1L16	119	0L86	1R125
23	0R23	1R24	120	0R126	1R127
24	0R25	1R26	121	0R58	0R110
25	0R27	1R28	122	0R99	1R113
26	0R29	1R30	123	0R79	1R78
27	0R31	1RH	124	0R88	1R88
28	0L32	1R33	125	0R128	1R129
29	0R34	1R35	126	0R32	1R36
30	0L36	1L37	127	0R31	1R111
31	0R38	1R39	128	0R130	1R131
32	0R40	1R41	129	0R132	1R133
33	0R42	1R43	130	0R134	1R135
34	0R44	1R45	131	0R136	1R137
35	0R46	1R33	132	0R138	1R139
36	0R47	1R48	133	0R140	1R141
37	0R49	1R50	134	0R97	1R35
38	0R51	1R42	135	0R142	1R124
39	0R52	1R53	136	0R143	1R144
40	0R54	1R55	137	0R145	1R146
41	0R56	1R57	138	0R134	1R147
42	0R58	0R59	139	0R88	1R148
43	0R60	0L14	140	0R149	1L131
44	0R61	1R62	141	0R150	1R151
45	0R63	1R52	142	0R76	1R41
46	0R64	1R65	143	0R28	1R152
47	0R58	0R66	144	0L153	1L154

q	read 0	read 1	q	read 0	read 1
48	0R67	0L14	145	0L129	1R155
49	0R68	0R69	146	0R156	1R157
50	0R70	0L48	147	0R149	1R158
51	0R64	1R53	148	0R159	1R88
52	0L71	0L15	149	0R97	1R160
53	0R72	0L10	150	0R161	1R162
54	0R73	1R74	151	0R163	0L10
55	0R75	1R76	152	0R164	1R165
56	0R77	1R78	153	0R166	1R157
57	0R79	1R80	154	0R167	1R156
58	0L81	0R1	155	0R135	1R130
59	0R82	1R59	156	0R147	1R41
60	0L73	1L46	157	0R134	1R168
61	0R83	1R84	158	0R41	1L169
62	0R54	1R85	159	0L170	1R171
63	0L86	1R87	160	0R107	1R85
64	0R58	0R82	161	0R172	1R170
65	0R88	1R30	162	0R173	1R174
66	0R89	1R66	163	0R175	1R176
67	0L90	1L33	164	0R177	1R178
68	0L81	0R2	165	0L179	1L180
69	0R91	1R69	166	0R149	1R140
70	0R72	1R92	167	0R165	1R181
71	0R46	1R93	168	0L130	1R88
72	0R42	0L13	169	0R182	1R135
73	0R94	1R37	170	0R183	1R145
74	0R30	1R95	171	0R158	1R130
75	0R42	1R96	172	0R139	1R165
76	0R97	1R98	173	0R158	1R184
77	0R99	1L100	174	0R185	1R107
78	0R30	1R77	175	0R31	1R52
79	0R93	1R101	176	0R186	1R175
80	0R30	1R102	177	0R134	1R142
81	1R81	1L15	178	0R32	1R187
82	0R103	1R82	179	0R144	1R188
83	0R94	0L7	180	0R189	1R190
84	0R104	1R105	181	0R41	1R191
85	0R42	1R106	182	0R168	1R41
86	0R107	1R48	183	0R192	1R88
87	0R108	0L72	184	0R153	1R146
88	0R68	0R66	185	0R165	1R180
89	0R109	1R89	186	0R92	1R111
90	0R49	1R100	187	0R193	1R159
91	0R110	1R91	188	0R169	1R154
92	0R30	1R48	189	0R158	1R192
93	0R47	1R60	190	0R179	1R182
94	0L63	0R89	191	0R182	1R41
95	0R94	1L111	192	0R149	1R168
96	0R101	0L14	193	0R76	1R144

B Archive contribution ledger

Archive	Contribution and interpretation
`Below_U208_Research_2026-10-29.zip`	Full-frame windows, offset quotients and guarded pruning; source 3,909→3,904; no accepted below-208 table.
`ConPA_below208_research_2026-10-29.zip`	Finite-model role-collapse objection, 279,936 fixed-association parameter cases, fold synthesis; no sub-208.
`PA195_ConPA_research_2026-11-14.zip`	Positive195S and195C: unchanged separate190 AX word, 67−6 readers; full source/runtime/startup certificates plus independent literal-tape audits.
`PA_consistency_TM_research_2026-11-13.zip`	159-node five-zero source and 157-node combined representation; exact source CFG checks, no smaller physical allocation.
`U208_ConPA_2026-10-29.zip`	Particular native172/211 history fold reduces a U209 parent to U208.
`U208_ConPA_Conditional_Context_2026-10-29.zip`	Conditional-entry proof; same two original U208 graph classes.
`U208_ConPA_Entry_Context_2026-10-29.zip`	Full entry-context proof; same two original classes.
`U208_ConPA_History_2026-10-29.zip`	Finite-history proof with four exceptional blocks; same original saving.
`conpa_tm_comparison_and_reduction_search.zip`	Experimental MP-result-in-H and MP-without-zero-push 196 variants; rejected Shift layouts halt at 43,563.
`conpa_tm_comparison_and_search.zip`	A/D register-swap 196; rejected two-rotation 195 fails 93 contracts.
`conpa_tm_study_2026-11-13.zip`	Combined196 = H-only Shift + MP-result-in-H: 3,788 sites, 4,928 contracts, 57,922 startup; direct-negation 197 too.
`pa194_conpa_2026-11-13(2).zip`	A further positive 194-state construction on the same separate190 source: 134 + 68 - 8, 32 simultaneous fold cases, 3,673 sites, 4,780 contracts, startup 58,154. Also retains a distinct separate195 seed; ordinary PA calibration, no new end-to-end kernel theorem.
`pa194_conpa_2026-11-13.zip`	Two positive 194-state constructions: primary 134 + 67 - 7 on the unchanged 195S source, and an A/D register allocation companion; crossed195 intermediate; independent literal-tape contracts and ordinary PA calibration; no new kernel-checked endpoint.
`pa195_conpa_2026-11-13.zip`	Positive original PA195: unchanged diagonal191 program, 68−7 reader allocation, detailed ordinary PA calibration.
`pa195_diagonal_combined_2026-11-13.zip`	PA195DC: original diagonal191 emitter plus cached-head Combined controller, 134 + 68 - 7 = 195, 3,676 sites, 4,784 contracts, startup 57,922; independent literal-tape audit. Two secondary195 logical variants pass finite checks (rho exchange and negative Gamma); additional calibration separately discussed.
`pa195_followup_research.zip`	Positive M195S8 uses 134 + 68 + 1 - 8 = 195; a rejected 194-state table has valid local folds but seven-zero source addresses break the scanner. Adds Gamma relocation, gate and witness results, and bounded negative searches.
`pa196_research_2026-11-13.zip`	Exact 157-node representation of 3,812-site original program; loses contextual sharing, no physical reduction.
`pa208_research_2026-10-29.zip`	Eighteen further helper layouts; 27 designated U208 presentations give 20 graphs, not two.

Archive	Contribution and interpretation
pa_below196_research.zip	Eight equivalent gates, membership/equality orientations, broad grammar searches; original201-token best 196.
pa_consistency_comparison_2026_11_13.zip	Modal-head/explosion split, twelve semantic compositions, joint gate; physical results 196–200.
pa_consistency_comparison_and_new196_2026-11-13.zip	Diagonal196 (3,700 sites), Separate196 (3,673), Scanner-best196; principal logical/source parents of positive195.
pa_consistency_cross_archive.zip	Four new196 combinations: diagonal/separate AX + Combined controller, each optionally zero-jump; sites 3,676/3,649 and startups57,922/58,168.
pa_consistency_reduction.zip	Rebracketed196, H-only196, integrated gate 203/206; two failed 195 diagnostics.
pa_consistency_reduction_audit_2026-11-14.zip	Four further experimental MP-H compositions, two196 and two197; parameter/order/association/helper/scanner negative searches.
pa_consistency_research.zip	Two exact158-node sources and repaired196; nominal195 needs added scanner state.
pa_consistency_research_2026-10-29.zip	Inductive 3,909→3,904 specialization; no physical sub208.
pa_consistency_research_2026-11-13(2).zip	Twelve-rotation196, diagonal198, successful-MP halting-source experiment; no sub196.
pa_consistency_research_2026-11-13.zip	Earlier diagonal-vacuity199 and short-negation197.
pa_consistency_research_2026-11-14.zip	NEG18: ‘INC7; INC7; node64‘, 18-site encoded negation; complete source3,441 and197 states (134+68−5); independent macro and scratch-zero proofs.
pa_tm_comparison_and_reductions.zip	Two rotated196 graphs; distributed-gate198 with 3,773 source sites.
pa_tm_extended_research_2026-11-13.zip	Four closing-decrement variants and stronger zero invariants; all change the same row, hence not stackable.
pa_tm_research_2026-11-13(2).zip	Zerojump196 (4,959 contracts, 58,402 startup); parity/disequality experiments; rejected195 fails source26.
pa_tm_research_2026-11-13.zip	Shorter-path196 and independent physical checking; detailed direct PA reverse-translation strategy.
pa_unary_power_cover.zip	Unary weak-covering theory and PA proof translations; no new compiled small table.
u208_comparison_research_2026-10-29.zip	Original two-class audit; weak/exact schemas; rejected207.
u208_research_2026-10-29.zip	Primary/alternative U208 and symbolic contracts; same initial history fold.

References

[1] ProveIt, *A 194-state Turing machine calibrated to Con(PA)*, `docs/incoming/pa194_conpa_2026-11-13.zip`, including `pa194_conpa/report/pa194_conpa.tex`, the decoder grammar, folds, and independent physical verification. Inspected at the pinned revision.

[2] ProveIt, `docs/incoming/PA195_ConPA_research_2026-11-14.zip`, members under `pa_consistency_research/construction195/`, `independent_verification/`, and `physical195_audit/`. The cached-head command audit and exact original construction are retained in `reference195C/`.

[3] ProveIt, `docs/incoming/pa194_conpa_2026-11-13(2).zip`, 194-state allocation with 68 raw readers and eight compatible folds; original separate-head controller. Included in the final revision refresh.

[4] ProveIt, `docs/incoming/pa195_diagonal_combined_2026-11-13.zip`, PA195DC: diagonal 191-token emitter with the cached-head controller. Included in the final revision refresh.

[5] ProveIt, `docs/incoming/pa195_followup_research.zip`, accepted M195S8 and the rejected seven-zero 194 quotient.

[6] ProveIt, *PA consistency machines*, pinned PA README.

[7] James Chang, *BusyBeaver*, PA196 snapshot `407cdc356f6b586635e71ac92be394255f4bfea4`, and PA209 snapshot `69d98241a721ead20b4f0b17df9062950b40fbfe`, as imported in ProveIt. The upstream MIT license and source provenance are retained.

[8] ProveIt, *Smaller machines for Con(ZFC)*, pinned ZFC README.

[9] ProveIt, *Concise presentations of intuitionistic primitive recursive arithmetic*, pinned concise-iPRA README.